# RAVI RACHAKONDA

*Staff Machine Learning Engineer · AI Platforms & Evaluation Infrastructure*

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## SUMMARY

Staff ML Engineer with 17+ years architecting and shipping AI systems at Amazon scale — from founding engineer on Alexa to key technical leader on Amazon Nova. Currently titled Senior MLE at Amazon while operating at Staff scope: setting technical direction across 4+ orgs, defining standards adopted by 10+ teams, and owning architecture decisions that directly shape Amazon's most strategic AI products. I build the platforms that decide whether a model or agent is ready: petabyte-scale data curation infrastructure, evaluation frameworks that gate launches, and managed platforms adopted by first- and third-party customers. Co-inventor on 5 granted US patents.

## SCALE AT A GLANCE

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|---|---|
| **Data & Training** | ▸ **17 trillion tokens**<br>▸ **Petabyte-scale multimodal datasets**<br>▸ **Several hundred machines** |
| **Production** | ▸ **1M+ image-generation requests / month**<br>▸ **10,000+ TPS at peak**<br>▸ **99.9% availability** |
| **Technical Leadership** | ▸ **Architecture adopted by 10+ internal & enterprise teams**<br>▸ **Cross-org alignment across 4+ organizations** |

## EXPERIENCE

### Senior Machine Learning Engineer, Nova Forge · Amazon AGI

*Jan 2026 – Present | Sunnyvale, CA*

- **Lead cross-organizational data infrastructure strategy** for Amazon Nova foundation models and Nova Forge, Amazon Bedrock's managed platform for foundation-model customization.
- **Own the curation-to-evaluation feedback loop** for Nova training data — connecting dataset deduplication, quality, and lineage decisions to measurable model-evaluation outcomes, so data changes are evaluated with the same rigor as model changes.
- **Own human-versus-model judge calibration** for offline evaluations — expert-rated anchor sets, agreement tracking over time, and judge-version pinning to keep results comparable across model and judge changes.
- **Defined the data-preparation architecture** adopted across 10+ internal and enterprise teams, standardizing customization workflows and accelerating customer onboarding.
- **Lead migration** from manual data-preparation workflows to a managed end-to-end platform, reducing customer time-to-first-trained-model.
- **Technical bridge between enterprise customers and platform teams** — translating customer requirements into platform roadmap decisions that shape Nova Forge's competitive positioning against OpenAI and Google fine-tuning offerings.

### Senior Research Engineer, Nova Pre & Post Training · Amazon AGI

*2024 – Jan 2026 | Sunnyvale, CA*

- **Designed and standardized the vision-language evaluation framework** used for Nova launch-readiness decisions, defining the metrics that gave leadership a consistent signal on model quality.
- **Cross-organizational technical lead** for Amazon Nova, a multimodal foundation-model family delivered to enterprise customers through Bedrock and competing with GPT-4o and Gemini.
- **Drove org-wide adoption of NVIDIA NeMo 2.0, MegatronCore, and MegatronEnergon** as the canonical large-scale training stack, consolidating tooling and reducing engineering overhead.
- **Reduced training-cluster MTTR by 13× (40 min → 3 min)** by redesigning the Energon dataloader, improving GPU utilization and reducing wasted compute.
- Designed and delivered the real-time Avatar demo for **AWS re:Invent 2024**, demonstrating Nova's multimodal capabilities.

### Senior Software Engineer / Tech Lead, Fire TV Background & Creative AI · Amazon

*2022 – 2024 | Sunnyvale, CA*

- **Founded the Image Generation initiative** spanning Alexa AI and Fire TV; shipped voice-driven background generation to millions of Fire TV devices as a differentiated premium feature.

- **Delivered the model productized as Amazon Bedrock Titan Image Generation (Nova)**, now processing 1M+ image-generation requests per month in production.
- **Defined architecture and technical standards** for an end-to-end MLOps pipeline with automated CI/CD, reducing deployment friction and establishing reusable engineering practices.
- **Founded Creative AI for Amazon Kids**, a 0→1 system generating personalized animated stories that ranked among the top three Amazon Kids Alexa skills at launch.

### Senior Software Engineer / Tech Lead, Alexa Astro · Amazon Lab126

*Aug 2018 – 2022 | Sunnyvale, CA*

- **Founded and led the Personality program for Amazon Astro**, defining the technical vision for expressive robot behavior and shipping the experience to ~500 devices at launch.
- **Established architecture and engineering standards** for the Personality system; patterns adopted across the Astro organization.

### Software Development Engineer / Tech Lead, Alexa Platform · Amazon Lab126

*Sep 2012 – Aug 2018 | United States*

- **Founding engineer on Amazon Alexa**; defined engineering patterns across Weather, Sports, Movies, Notifications, and Spatial Perception as Alexa scaled to hundreds of millions of devices.
- **Tech lead for Echo Spatial Perception (2017–18)**; co-authored patents supporting multi-device arbitration, a core Alexa capability.
- **Tech lead for Alexa Notifications (2016–17)**; architected the notification system enabling first- and third-party skills to reach users across all Alexa devices.
- **Scaled Alexa platform services to 10,000+ TPS** at 99.9% availability during peak holiday traffic.

## TECHNICAL SKILLS

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|---|---|
| **AI & ML** | LLM-as-judge evaluation, agentic trace evaluation, PyTorch, vLLM, NVIDIA NeMo 2.0, MegatronCore, MegatronLM, MegatronEnergon, Triton Inference Server, diffusion transformers, multimodal LLMs, vision-language models |
| **Data & Distributed Systems** | Ray, Ray Data, Apache Spark, EMR, Apache Iceberg, FAISS, MinHash LSH, distributed training, data parallelism, fault-tolerant pipelines at petabyte scale |
| **Cloud & MLOps** | SageMaker, SageMaker HyperPod, AWS Batch, ECR, S3, Lambda, DynamoDB, EC2, CloudWatch, CI/CD |
| **Languages & Frameworks** | Python, Java, JavaScript, C++ (intermediate), Perl, Shell, FastAPI, React, Jupyter, Alexa Skills Kit |

## PATENTS

- **Context-Based Device Arbitration |** US #10546583 · US #11289087 · US #12033632
- **Device Selection from Audio Data |** US #10685669 · US #11600291

## EDUCATION

**B.Tech, Information Technology** · Indian Institute of Information Technology, Prayagraj, India · 2003 – 2007